Manik Bansal

Postdoctoral Researcher

Injury Biomechanics Laboratory,

Department of Engineering, Robert Morris University, Pittsburgh, PA, US

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Nationality: Indian | | **US Visa** : H1-B (Since Jan25) | **Preferred joining Date**: Sept 2026 to Jan 2027

Education

Indian Institute of Technology (IIT) Roorkee — Roorkee, UK, India
PhD in Mechanical Engineering (Design) — Dec. 2014 - Sept. 2019

Vellore Institute of Technology — Vellore, TN, India
M.Tech in CAD/CAM (Mechanical Engg.) — July. 2011 - June. 2013

Singhad Academy of Engineering, University of Pune — Pune, MH, India
B.E. in Mechanical Engineering — July. 2005 - June. 2009

Areas of Interest

- Computational Solid Mechanics
- Multiscale Finite Element Modeling
- Fracture and Fatigue Mechanics
- Injury Biomechanics and Soft-Tissue Mechanics
- High-Performance Scientific Computing (HPC)

Skills

Scientific Programming	MATLAB, Python, TCL script, Shell Scripting
High-Performance Computing	Linux HPC Clusters, SLURM scripting, PSC Bridges-2
Modeling & Simulation Techniques	Extended FEM (XFEM), Multiscale Modeling, Arbitrary Eulerian-Lagrangian (AEL) Methods, Quasi-Static Modeling, Explicit/Implicit Dynamics, Fracture/Damage Modeling
CAD Tools	Solidwords, Creo, AutoDesk MeshMixer, Onshape
CAE Tools	LSDyna, ABAQUS , ANSYS, FeBio
Medical Image Processing	3D Slicer, Avizo, ImageJ, HyperMesh, CUBIT, MATLAB Gibbon ToolBox
Scientific Visualization	ParaView, HyperView, Origin, MATLAB Visualization
Documentation Tools	LaTeX, Origin, Microsoft Word/Excel/Powerpoint, Github
Experimental Mechanics	Fracture Toughness Test (MTS), Tension Test (MTS, Instron)
Experimental Biomechanics	Micro-CT Imaging, DIC, Soft-Tissue Testing, Pressure Inflation Testing
Biological Tissue Handling	Animal Tissue Dissection (Pig: Brain & Eye, Goat: Eye), Tissue Fixation

Professional Experience

Teaching: 2+ years, Research: 10+ years, Industry: 1 year

Postdoctoral Researcher - Robert Morris University — Pittsburgh, PA, US
Department of Engineering, School of Engineering, Mathematics and Science — Nov. 2022 - Present
Developing and optimizing high fidelity human head models for predicting blast induced traumatic brain injury in human brain.

Part-Time Faculty - Robert Morris University — Pittsburgh, PA, US
Department of Engineering, School of Engineering, Mathematics and Science — Jan. 2024- May 2025
Teaching biomedical and biomechanics engineering subjects to undergraduate students.

Postdoctoral Associate - University of Pittsburgh — Pittsburgh, PA, US
Department of Ophthalmology, School of Medicine — Feb. 2020 - Oct. 2022
Developed micro-mechanics based numerical models to determine the influence of collagen fiber orientation, interweaving and crimp in sclera on mechanics of eye.

Senior Research Fellow - Indian Institute of Technology (IIT) Roorkee — Roorkee, UK, India
Mechanical and Industrial Engineering Department — Nov. 2014 - Sept. 2019
Developed computationally efficient XFEM multi-scale framework for the uncertainty quatification of heterogeneous materials with applications in the energy systems.

Assistant Professor - Lovely Professional University — Phagwara, PB, India
School of Mechanical Engineering — Aug. 2013 - Oct. 2014
Taught mechanical design engineering subjects. Mentored both graduate and under-grad students.

Assistant Project Engineer - Sakshi Innovation Pvt.Ltd — Ludhiana, PB, India
Department of Machine Tool Design & Research — Aug. 2009-Nov. 2010
Designed and supervised fabrication of special purpose machine tools for Lightweight Expanded Clay Aggregate (LECA) plant.

Journal Publications

[Current research-impact metrics (source: Google Scholar): Citations-157, h-Index – 7, i10-index – 6]

Published

- **Manik Bansal**, Bingrui Wang, Susannah Waxman, Fuqiang Zhong, Yi Hua, Yuankai Lu, Juan Reynaud, Brad Fortune, and Ian A. Sigal. "Proposing a methodology for axon-centric analysis of IOP-induced mechanical insult." ***Investigative Ophthalmology & Visual Science*** *65 (13) (2024)*. [Link] [Acceptance rate: 31.5%, H-index: 260]
- Islam, Mohammad R., Fengting Ji, **Manik Bansal**, Yi Hua, and Ian A. Sigal, "Fibrous Finite Element Modeling of the Optic Nerve Head Region.", ***Acta biomaterialia*** *175 (2024)* [Link] [H-index: 261]
- Fengting Ji, **Manik Bansal**, Bingrui Wang, Yi Hua, Felix Matuschke, Markus Axer and Ian A. Sigal, "A direct fiber approach to model sclera collagen architecture and biomechanics", ***Experimental Eye Research*** 232 (2023): 109510. [Link] [H-index: 153]
- **Manik Bansal**, Subrato Sarkar and I. V. Singh, "An XFEM-strain gradient damage model for efficient modeling of materials with reinforcement particles", ***Engineering Fracture Mechanics*** 271 (2022): 108667. [Link] [Acceptance rate: 30%, H-index: 166]
- **Manik Bansal**, I. V. Singh, R. U. Patil, Susanne Claus and S.P.A. Bordas, "A Simple and Robust Computational Homogenization Approach for Heterogeneous Particulate Composites" ***Computer Methods in Applied Mechanics and Engineering*** 349 (2019): 45-90. [Link] [Acceptance rate: 19%, H-index: 246]
- **Manik Bansal**, I. V. Singh, B. K. Mishra and S.P.A. Bordas, "A Parallel and Efficient Multi-Split XFEM for 3-D Analysis of Heterogeneous Materials", ***Computer Methods in Applied Mechanics and Engineering*** 347 (2019): 365-401. [Link] [Acceptance rate: 19%, H-index: 246]
- **Manik Bansal**, I. V. Singh, B. K. Mishra, Kamal Sharma and I. A. Khan, "A two-scale stochastic framework for predicting failure strength probability of heterogeneous materials", ***Composite Structures*** 179 (2017): 294-325. [Link] [Acceptance rate: 15%, H-index: 213]
- **Manik Bansal**, I. V. Singh, B. K. Mishra, Kamal Sharma and I. A. Khan, "A stochastic XFEM model for the tensile strength prediction of heterogeneous graphite based on microstructural observations", ***Journal of Nuclear Materials*** 487 (2017): 143-157. [Link] [Acceptance rate: 42%, H-index: 174]
- **Manik Bansal**, I. V. Singh, B. K. Mishra, Kamal Sharma, and I. A. Khan, "A numerical prediction of flexural strength probability for NBG-18 nuclear grade graphite using strength pair model", ***The Journal of Strain Analysis for Engineering Design*** 52, no. 3 (2017): 204-211. [Link] [Acceptance rate: 33.3%, H-index: 60]

In Progress

- **Manik Bansal**, Baudouin Fonkwa, Alice Lux Fawzi, Eric Johnsen, Christian Franck and, Rika Wright Carlsen, "Microcavitation as potential brain injury mechanism in repetitive blast exposure.", ***Journal: PNAS***[Submission Pending: First Draft] [Link]

Service

Conference Abstract Reviewer
- Summer Biomechanics, Bioengineering and Biotransport Conference (SB3C): 2022, 2025
- American Society of Biomechanics (ASB): 2025

Conference Paper Reviewer
- ASME International Mechanical Engineering Congress & Exposition (IMECE): 2026 (2 papers- *Ongoing*)

Journal Article Reviewer
- *International Journal of Damage Mechanics* (2020)
- *Engineering Structures* (2022, 2024)
- *Applied Composite Materials* (2024, 2025)
- *Experimental Eye Research* (2025)
- *Progress in Additive Manufacturing* (2025)
- *Military Medicine* (2025, 2026–*Ongoing*)
- *Frontiers in Bioengineering* (2026–*Ongoing*)

Co-Moderator
- Acoustical Society of America (ASA) Webinar: *"Cavitation-Induced Brain Injury: Mechanisms and Modeling"* (Nov. 20, 2025)

Professional Teaching & Mentoring Experience

Robert Morris University, Pittsburgh, PA, USA [Position: Postdoctoral Researcher]

Semester	Course	Details
Spring 2025	ENGR 2510A	Biomedical Engineering Principles (*Instructor*)
Spring 2025	ENGR 4900	Mentored undergraduate student Marcus T. Catlett on the project: *Design and Fabrication of a Tension Device for In-Situ Analysis of Optic Nerve Damage using Micro-CT Imaging*
Spring 2024	ENGR 2510A	Biomedical Engineering Principles (*Co-Instructor*)
Spring 2024	ENGR 4900	Mentored a student design team on the project: *Design and Fabrication of a 3D BioPrinter* (Team lead: Madeline Kinton)

University of Pittsburgh, Pittsburgh, PA, USA[Position: Postdoctoral Researcher]

Semester	Course	Details
Fall 2020	BIOSC 1901	Mentored undergraduate Biology student Sinead E. Healy on a computational mini-project: *Visualization of Retinal Ganglion Cell Axon Pathways through the Optic Nerve Head*
Spring 2021	BIOSC 1903	Mentored undergraduate Biology student Sinead E. Healy on a computational mini-project: *User-Friendly Fiber-Based Model Reconstruction*

Indian Institute of Technology Roorkee, India [Position: Senior Research Fellow (PhD)]

Semester	Course	Details
Fall 2018	MIN 554	Computer Aided Mechanism Design (*Teaching Assistant*)

Lovely Professional University (LPU), Punjab, India [Position: Assistant Professor]

Semester	Course	Details
Fall 2013	MEC 533	Industrial Tribology
Fall 2013	MEC 320	Advanced Creo Modeling (CAD Lab)
Fall 2013	MEC 223	Machine Design-I
Spring 2014	MEC 201	Mechanics of Materials
Spring 2014	MEC 423	Finite Element Method
Spring 2014	MEC 494	Mentored undergraduate capstone project: *Three-Point Bending Setup for Measuring Composite Material Strength* (Student: Prabhat Mishra)
Spring 2014	MEC 494	Mentored undergraduate design team project: *Design and Fabrication of an All-Terrain Vehicle (ATV)* (Team lead: Aman Sharma)
Fall 2014	MEC 538	Applied Elasticity and Plasticity
Fall 2014	MEC 641	Advanced Strength of Materials
Fall 2014	MEC 652	Design of Pressure Vessel and Piping

Academic Achievements, Fellowships/Scholarships & Awards

2025	**Recognition Award: Completion of Department of Defence-University of Wisconsin, Grant Funded Project 2020-23**, 9th Annual Research and Grant Expo, Robert Morris University	Pittsburgh-PA, United States
2025	**Recognition Award: Completion of Department of Defence-University of Pittsburgh, Grant Funded Project 2020-24**, 9th Annual Research and Grant Expo, Robert Morris University	Pittsburgh-PA, United States
2017-19	**MHRD Fellowship (₹840,000)**, Indian Institute of Technology Roorkee	Roorkee, India
2014-17	**Senior Research Fellowship (₹828,266)**, Indian Institute of Technology Roorkee	Roorkee, India
2013	**2nd Overall Rank**, M.Tech CAD/CAM Degree Convocation, Vellore Institute of Technology	Vellore, India
2012	**Merit Scholarship (Best Academic Performance Award)**, University Day, Vellore Institute of Technology	Vellore, India
2011	**Merit Scholarship (Best Academic Performance Award)**, University Day, Vellore Institute of Technology	Vellore, India
2011	**3rd Place**, Oral Presentation- SET Conference, Vellore Institute of Technology	Vellore, India

Conference Posters and Talks

Presenter underlined

2026

Podium **Manik Bansal**, Baudouin Fonkwa, Alice Lux Fawzi, Eric Johnsen, Christian Franck, and Rika Wright Carlsen, "Subject-Specific Human Digital Twin for Predicting Blast-Induced Traumatic Brain Injury," *20th U.S. National Congress on Theoretical and Applied Mechanics*, Pasadena, CA, USA, June 21–25, 2026. *Upcoming*

Podium Shuvo Chowdhury, **Manik Bansal**, Rika Carlsen, and Ashfaq Adnan, "Morphology-informed Hyper-viscoelastic Micromechanical Response of Neocortex," *Society of Engineering Science (SES) Annual Technical Meeting*, Purdue University, West Lafayette, IN, USA, Oct. 11–14, 2026. *Upcoming*

Podium Enze Chen, Arvin Chen, Aashutosh Kulakarni Prachet, Cody Ottinger, William Jackson, Sarah Bentil, **Manik Bansal**, Showren Datta, Rika Carlsen, Jing Zhang, Haneesh Kesari, and Christian Franck, "Blast-induced Large Deformations in Brain-like Soft Matter," *Society of Engineering Science (SES) Annual Technical Meeting*, Purdue University, West Lafayette, IN, USA, Oct. 11–14, 2026. *Upcoming*

Podium Enze Chen, Arvin Chen, Aashutosh Kulakarni Prachet, Cody Ottinger, William Jackson, Sarah Bentil, **Manik Bansal**, Rika Carlsen, Jing Zhang, Haneesh Kesari, and Christian Franck, "Blast-induced Large Deformations in Brain-like Soft Matter," *20th U.S. National Congress on Theoretical and Applied Mechanics*, Pasadena, CA, USA, June 21–25, 2026. *Upcoming*

Poster **Manik Bansal**, Baudouin Fonkwa, Eric Johnsen, Christian Franck, and Rika Wright Carlsen, "A Computationally Efficient Scheme for Predicting Blast-Induced Traumatic Brain Injury," *10th Annual Research and Grant Expo*, Robert Morris University, Pittsburgh, PA, USA, March 26, 2026.

Podium **Manik Bansal**, Baudouin Fonkwa, Eric Johnsen, Alice Lux Fawzi, Haneesh Kesari, Christian Franck, and Rika Wright Carlsen, "From Head Loading to Injury Risk: Computational Modeling of Brain Injury Risk in Military Service Members," *23rd Annual NeuroTech Convention of the Society for Brain Mapping and Therapeutics (SBMT)*, Los Angeles, CA, USA, April 2026.

2025

Podium **Manik Bansal**, Baudouin Fonkwa, Alice Lux Fawzi, Eric Johnsen, Christian Franck, and Rika Wright Carlsen, "Computational Modeling of Blast-Induced Traumatic Brain Injury from Repetitive Blast Exposure," *9th International Forum on Blast Injury Countermeasures (IFBIC)*, Tokyo, Japan, May 7–9, 2025.

Podium **Manik Bansal**, Baudouin Fonkwa, Eric Johnsen, Alice Lux Fawzi, Christian Franck, and Rika Wright Carlsen, "Quantifying Brain Injury Risk for Subconcussive Blast Exposures," *22nd Annual NeuroTech Convention of the Society for Brain Mapping and Therapeutics (SBMT)*, Los Angeles, CA, USA, Feb. 27–March 2, 2025.

Poster **Manik Bansal** and Rika Wright Carlsen, "Investigating Impact of Varying Fidelity in Human Head Models on Blast-Induced Traumatic Brain Injury," *48th American Society of Biomechanics Annual Meeting*, Pittsburgh, PA, USA, Aug. 13–16, 2025.

Poster **Manik Bansal** and Rika Wright Carlsen, "Investigating Impact of Varying Fidelity in Human Head Models on Blast-Induced Traumatic Brain Injury," *PANTHER TBI Workshop*, Madison, WI, USA, Sept. 15–17, 2025.

Poster **Manik Bansal**, Baudouin Fonkwa, Eric Johnsen, Christian Franck, and Rika Carlsen, "Numerical Modelling of Repetitive Low-Level Blast Scenarios to Quantify Likelihood of Cavitation-Induced Traumatic Brain Injury," *9th Annual Research and Grant Expo*, Robert Morris University, Pittsburgh, PA, USA, March 26, 2025.

Poster Marcus T. Catlett, **Manik Bansal**, Devin R. E. Cortes, Walter Schneider, and Rika Wright Carlsen, "Development of Tension Device for In-Situ Assessment of Optic Nerve Damage using Micro-CT Imaging," *BioE Day*, University of Pittsburgh, Pittsburgh, PA, USA, March 20, 2025.

2024

Podium **Manik Bansal**, Baudouin Fonkwa, Eric Johnsen, Alice Lux Fawzi, Christian Franck, and Rika Wright Carlsen, "Quantifying Blast Safety Limits to Prevent Sub-Concussive Traumatic Brain Injury During Heavy Weapons Training," *Military Health System Research Symposium (MHSRS)*, Kissimmee, FL, USA, Aug. 26–29, 2024.

Poster **Manik Bansal**, Baudouin Fonkwa, Eric Johnsen, Christian Franck, and Rika Carlsen, "Numerical Modelling of Low-Level Blast Scenarios to Quantify Likelihood of Cavitation-Induced Traumatic Brain Injury," *Summer Biomechanics, Bioengineering and Biotransport Conference*, Lake Geneva, WI, USA, June 11–14, 2024.

Poster **Manik Bansal**, Baudouin Fonkwa, Eric Johnsen, Christian Franck, and Rika Carlsen, "Numerical Modelling of Low-Level Blast Scenarios to Quantify Likelihood of Cavitation-Induced Traumatic Brain Injury," *PANTHER TBI Workshop*, Madison, WI, USA, Sept. 15–17, 2024.

Poster William White, Baudouin Fonkwa, **Manik Bansal**, Rika Carlsen, Alice Lux Fawzi, Christian Franck, and Eric Johnsen, "A Wave-Based Framework for Understanding Cavitation Damage Potential in Blast-Induced Traumatic Brain Injury," *PANTHER TBI Workshop*, Madison, WI, USA, Sept. 15–17, 2024.

2023

Podium **Manik Bansal**, Baudouin Fonkwa, Eric Johnsen, Rika Carlsen, and Christian Franck, "Investigating Cavitation-Induced Primary Injury during Open-Field Blast to Determine Safety Limits for the Warfighter," *Military Health System Research Symposium (MHSRS)*, Kissimmee, FL, USA, Aug. 14–17, 2023.

Poster **Manik Bansal**, Benjamin E. Grossman-Ponemon, Anu Tripathi, Haneesh Kesari, and Rika Carlsen, "Computationally Efficient Framework to Quantify Head Kinematics Induced Brain Injury from Open-Field Blast," *Military Health System Research Symposium (MHSRS)*, Kissimmee, FL, USA, Aug. 14–17, 2023.

Poster **Manik Bansal**, Baudouin Fonkwa, Eric Johnsen, Rika Carlsen, and Christian Franck, "Investigating Cavitation-Induced Primary Injury during Open-Field Blast to Determine Safety Limits for the Warfighter," *PANTHER TBI Workshop*, Madison, WI, USA, Sept. 19–21, 2023.

Podium Bingrui Wang, **Manik Bansal**, Fuqiang Zhong, Yi Hua, Juan Reynaud, Brad Fortune, and Ian Sigal, "Longitudinal Axon Compression as a Potential Mechanism Relating IOP-Induced Lamina Cribrosa Distortion and Glaucomatous Neural Tissue Loss," *Investigative Ophthalmology & Visual Science*, 64(8): 811, New Orleans, LA, USA, April 23–27, 2023. Link

2022

Podium **Manik Bansal**, Fuqiang Zhong, Yi Hua, Bingrui Wang, Juan Reynaud, Brad Fortune, and Ian Sigal, "A Novel Method to Determine Axon-Level IOP-Induced Insult," *Investigative Ophthalmology & Visual Science*, 63(7): 811, Denver, CO, USA, May 1–4, 2022. Link

Poster **Manik Bansal**, Fuqiang Zhong, Yi Hua, Juan Reynaud, Brad Fortune, and Ian Sigal, "Estimation of Insult to Retinal Ganglion Cell Axons Induced by Elevated Intraocular Pressure: An Axon-Centric Approach," *15th Postdoctoral Data & Dine Symposium*, Pittsburgh, PA, USA, May 25, 2022.

Podium Ian A. Sigal, **Manik Bansal**, Fuqiang Zhong, Yi Hua, Juan Reynaud, and Brad Fortune, "The Strain that Broke the Axon's Back: Towards Axon-Centric Optic Nerve Head Biomechanics," *ISER/BrightFocus Glaucoma Symposium*, Atlanta, GA, USA, May 24–27, 2022.

Poster Fengting Ji, Marissa Quinn, **Manik Bansal**, Yi Hua, and Ian A. Sigal, "Corneo-Sclera Collagen Fibers Exhibit Substantial Transverse Inclination," *Investigative Ophthalmology & Visual Science*, 63(7): 2719-A0083, Denver, CO, USA, May 1–4, 2022. Link

Podium Mohammad R. Islam, Fengting Ji, **Manik Bansal**, Yi Hua, and Ian A. Sigal, "A Novel Fibrous Finite Element Model of Soft Tissues," *Summer Biomechanics, Bioengineering and Biotransport Conference*, MD, USA, June 20–23, 2022.

2021

Poster **Manik Bansal**, Fengting Ji, Yi Hua, Fuqiang Zhong, and Ian A. Sigal, "Stress and Strain Distribution Analysis of Specimen-Specific Interweaved Fiber Sclera Model under Equi-Biaxial Strain," *Summer Biomechanics, Bioengineering and Biotransport Conference (SB3C)*, Virtual, June 14–18, 2021.

Poster Fengting Ji, **Manik Bansal**, Bingrui Wang, Yi Hua, and Ian A. Sigal, "Mechanical Properties of Scleral Collagen Fibers Obtained Using a New Fiber-Based Specimen-Specific Model of Sclera Microstructure," *Investigative Ophthalmology & Visual Science*, 62(8): 1652, Virtual, Sept. 24–26, 2021. Link

Podium Fuqiang Zhong, Yi Hua, Junchao Wei, **Manik Bansal**, Juan Reynaud, Brad Fortune, and Ian A. Sigal, "In-Vivo Evidence of Increased Lamina Cribrosa Compliance at Onset of Experimental Glaucoma in Nonhuman Primates," *Investigative Ophthalmology & Visual Science*, 62(8): 1827, Virtual, Sept. 24–26, 2021. Link

Podium Ian A. Sigal, Fuqiang Zhong, Yi Hua, Junchao Wei, **Manik Bansal**, Juan Reynaud, and Brad Fortune, "In-Vivo Evidence of Lamina Cribrosa Deformations Induced by Chronically Elevated IOP in Non-Human Primates," *Investigative Ophthalmology & Visual Science*, 62(8): 1862, Virtual, Sept. 24–26, 2021. Link

Poster Fengting Ji, **Manik Bansal**, Bingrui Wang, Yi Hua, Felix Matuschke, Markus Axer, and Ian A. Sigal, "Specimen-Specific Fiber-Based Modeling and Simulation of Sclera Collagen Fiber Microstructure and Biomechanics," *Summer Biomechanics, Bioengineering and Biotransport Conference (SB3C)*, Virtual, June 14–18, 2021.

Poster Fuqiang Zhong, Yi Hua, Junchao Wei, **Manik Bansal**, Juan Reynaud, Brad Fortune, Ziyi Zhu, and Ian A. Sigal, "Lamina Cribrosa Deformation Induced In-Vivo by Acutely and Chronically Elevated Intraocular Pressure," *Summer Biomechanics, Bioengineering and Biotransport Conference (SB3C)*, Virtual, June 14–18, 2021.

Podium Fuqiang Zhong, Yi Hua, Junchao Wei, **Manik Bansal**, Juan Reynaud, Brad Fortune, and Ian A. Sigal, "A High-Accuracy Digital Volume Correlation of Soft Tissue Biomechanics Characterization Using Optical Coherence Tomography," *Summer Biomechanics, Bioengineering and Biotransport Conference (SB3C)*, Virtual, June 14–18, 2021.

2012-16

Podium **Manik Bansal**, I. V. Singh, B. K. Mishra, and Kamal Sharma, "Numerical Evaluation of Flexural Strength of NBG-18 Graphite," *6th International Congress on Computational Mechanics and Simulation Materials (ICCMS)*, IIT Bombay, India, June 27–July 1, 2016.

Podium **Manik Bansal**, R. Pratap, Y. Bisht, I. V. Singh, B. K. Mishra, Kamal Sharma, and I. A. Khan, "Strength Evaluation of Nuclear Graphite using Semi-Homogenized XFEM and Modulus Degradation Approach," *60th Congress of ISTAM*, MNIT Jaipur, India, Dec. 16–19, 2015.

Podium **Manik Bansal**, R. Vasudevan, N. Manickam, and Manoj Buragohain, "Numerical Investigation of Circumferential Delamination in a Laminated Composite Cylindrical Shell under Internal Pressure," *International Conference on Energy Efficient Technologies for Sustainability (ICEETS)*, Nagercoil, India, April 10–16, 2013. IEEE Catalog No. CFP1375U-PRT, ISBN: 978-1-4673-6147-7.

Proceedings S. N. S. H. Chittajallu, **Manik Bansal**, A. R. Bedare, and Sunil Bhat, "Effect of Plane Stress and Plane Strain Conditions over Fracture Parameters of Ductile Plate in SSY (LEFM) and EPFM Regimes," *Advances in Civil Engineering and Building Materials*, pp. 239–242, Nov. 2012.

Grant-Funded Research Experience

Office of Naval Research (ONR) – PANTHER Program

Position Postdoctoral Researcher
Duration Oct 2022 – Present
Project Multiscale, physics-based approaches for blast and blunt traumatic brain injury prediction and prevention
Grant Nos. N00014-21-1-2855, N00014-21-1-2044, N00014-24-1-2200, N00014-22-1-2828
Contributions
- Developed high-fidelity computational frameworks for blast traumatic brain injury (bTBI) biomechanics.
- Performed multiscale finite element simulations of open-field blast loading.
- Contributed to injury threshold analysis, manuscripts, and annual technical reports.

Department of Defense (DoD) – Vision Research Program

Position Postdoctoral Researcher
Duration Oct 2023 – Present
Project High-Definition Axonal and Connective Tissue Imaging (HD-ACTI) in traumatic optic nerve injury
Grant No. VR190139
Contributions
- Designed in-situ micro-CT compatible tension devices for optic nerve injury imaging.
- Conducted ex-vivo porcine optic nerve mechanical testing.
- Developed computational models for traumatic optic nerve injury biomechanics.
- Mentored undergraduate engineering design teams associated with the project.

National Science Foundation (NSF)

Position Postdoctoral Researcher
Duration Jan 2022 – Oct 2022
Project Managing Glaucoma in the Digital Age: A New Tonometer to Connect Patients to their Caregivers
Grant No. NSF-2014389
Contributions
- Developed and optimized experimental frameworks for HNSW-based tonometry systems.
- Evaluated ex-vivo corneal pressure prediction methodologies.
- Supported manuscript preparation and technical documentation.

National Institutes of Health (NIH/NEI)

Position Postdoctoral Researcher
Duration Feb 2020 – Oct 2022
Project Optic nerve head microstructure, biomechanics, and susceptibility to glaucoma
Grant No. R01EY023966
Contributions
- Developed microstructure-based constitutive and finite element models of ocular tissues.
- Investigated axon-specific biomechanical insult mechanisms under elevated intraocular pressure.
- Contributed to computational workflow development, manuscripts, and grant-support activities.

Council of Scientific and Industrial Research (CSIR), India

Position PhD Researcher
Duration May 2018 – Sept 2019
Project Stochastic multiscale frameworks for heterogeneous materials
Grant No. 22(0777)/18/EMR-II
Contributions
- Developed stochastic multiscale computational frameworks for heterogeneous materials.
- Implemented MATLAB-based multiscale simulations and experimental validation workflows.
- Contributed to journal publications and technical presentations.

Board of Research in Nuclear Sciences (BRNS-BARC), India

Position	PhD Researcher
Duration	Nov 2014 – Aug 2017
Project	Prediction of graphite failure strength using RVE and XFEM approaches
Grant No.	36(2)/14/01/2014-BRNS/1078
Contributions	• Developed XFEM-based computational frameworks for stochastic fracture prediction. • Performed numerical implementation and experimental validation in MATLAB. • Contributed to technical publications and annual project reports.